

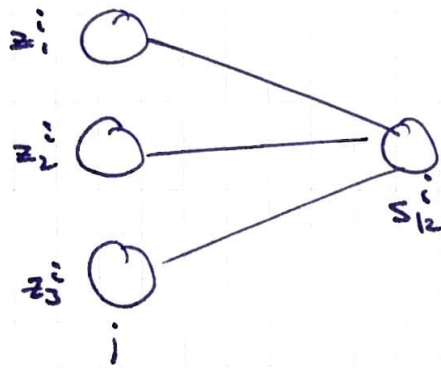
GLOROT INITIALISATION

The UNACTIVATED OUTPUT IS GIVEN BY:

$$s^i = z^i w^i + b^i$$

Thus for each unit s_k^i , we index on the weights

$$s_k^i = \sum_l z_l^i w_{l,k}^i + b_k^i$$



Now, since $z_k^{i+1} = f(s_k^i)$

$$s_j^{i+1} = \sum_k z_k^{i+1} w_{k,j}^{i+1} + b_j^{i+1}$$

$$s_j^{i+1} = \sum_k f(s_k^i) w_{k,j}^{i+1} + b_j^{i+1}$$

We can now setup the chain rule given layer $i+1$.

$$\begin{aligned} \frac{\partial \text{Cost}}{\partial s_k^i} &= \sum_j \frac{\partial \text{Cost}}{\partial s_j^{i+1}} \frac{\partial s_j^{i+1}}{\partial s_k^i} \rightarrow \frac{\partial s_j^{i+1}}{\partial s_k^i} = f'(s_k^i) w_{k,j}^{i+1} \\ &= f'(s_k^i) \sum_j w_{k,j}^{i+1} \frac{\partial \text{Cost}}{\partial s_j^{i+1}} \end{aligned}$$

which can be expressed as a dot product.

$$= f'(s_k^i) w_{k,\cdot}^{i+1} \cdot \frac{\partial \text{Cost}}{\partial s^{i+1}}$$

So this is just the gradient of one neuron calculated through the gradient of the next layer.
Now we must find the gradient of the weight.

$$\frac{\partial \text{Cost}}{\partial w_{l,k}^i} = \frac{\partial \text{Cost}}{\partial s_k^i} \frac{\partial s_k^i}{\partial w_{l,k}^i}$$

$$= z_l^i \frac{\partial \text{Cost}}{\partial s_k^i} \rightarrow \text{We computed this earlier.}$$

Now, we appropriate under the assumption of the hyperbolic tangent. Hyperbolic tangents are linear near 0, thus we assume:

$$f(s) \approx s \quad f'(s) = 1$$

Furthermore, we assume $E(s_k^i) = 0$, and the weights in some layer i share a variance $\text{Var}[w^i]$

WE FIRST CONSIDER THE FORWARD PASS.

$$s_k^i = \sum_{l=1}^{n_i} z_l^i w_{l,k}^i \quad \text{Var}[s_k^i] = \sum_{l=1}^{n_i} \text{Var}[z_l^i w_{l,k}^i]$$

$$= \sum_{l=1}^{n_i} \text{Var}[z_l^i] \text{Var}[w_{l,k}^i]$$

$$\text{Var}[s^i] = n_i \text{Var}[w^i] \text{Var}[z^i]$$

IN LINEAR ASSUMPTIONS, we have $z^{i+1} = s^i$, thus each layer multiplies through $n_i \text{Var}[w^i]$. We can then compound the variance from the first layers:

$$\text{Var}[z^i] = \text{Var}[x] \prod_{i=1}^{i-1} n_i \text{Var}[w^i].$$

This can be derived from below:

$$\begin{aligned} \text{Var}[z^{i+1}] &= \text{Var}[s^i] = n_i \text{Var}[w^i] \text{Var}[z^i] \\ &= n_i \text{Var}[w^i] [n_{i-1} \text{Var}[w^{i-1}] \text{Var}[z^{i-1}]] \\ &= \dots \\ &\stackrel{\text{Input}}{=} \text{Var}[x] \prod_{i=1}^{i-1} n_i \text{Var}[w^i]. \end{aligned}$$

IN PLAIN WORDS, THE VARIANCE OF THE FORWARD PASS IS THE VARIANCE OF THE INPUT MULTIPLIED BY THE COMPOUNDED WEIGHT VARIANCE.

NOW, CONSIDER THE BACKWARD PASS.

We notate the gradient at a layer i and neuron k as:

$$g_k^i = \frac{\partial \text{Cost}}{\partial s_k^i}$$

Layer i has n_i units, and weight matrix W has dimension $n_i \times n_{i+1}$. Thus

- s^i & z^{i+1} both have n_{i+1} entries.
- W^{i+1} is $n_{i+1} \times n_{i+2}$, so row k of W^{i+1} has n_{i+2} entries.

So if we pass down the gradient from the forward layer we get

$$g_k^i = f'(s_k^i) \sum_{j=1}^{n_{i+2}} W_{k,j}^{i+1} g_j^{i+1}$$

But the linear regime lets $f'(s_k^i) = 1$, so

$$g_k^i = \sum_{j=1}^{n_{i+2}} W_{k,j}^{i+1} g_j^{i+1}.$$

The terms are uncorrelated thus for layer i we can show

$$\text{Var}[g^i] = n_{i+2} \text{Var}[W^{i+1}] \text{Var}[g^{i+1}].$$

Similar to the forward pass, we expand

$$= n_{i+2} \text{Var}[W^{i+1}] (n_{i+3} \text{Var}[W^{i+2}] \text{Var}[g^{i+2}])$$

Thus if the top layer is d .

$$\text{Var}[g^i] = \text{Var}[g^d] \prod_{i=i+1}^d n_{i+1} \text{Var}[W^i].$$

Similarly to the forward pass, the variance of the gradient is the gradient of the last layer multiplied by the compounded weight variance.

OUR GOAL IS TO MAKE THE VARIANCE IN THE FORWARD AND BACKWARD PASS STABLE.

IN BOTH THE FORWARD AND BACKWARD TERMS THE VARIANCE IS MULTIPLIED BY THE TERM $n_i \text{Var}[w_i]$ & $n_{i+1} \text{Var}[w_i]$.

THUS; IF BOTH ARE 1 THEN BOTH FORWARD & BACKWARD PASSES STAY STABLE.

$$\therefore \text{Var}[w_i] = \frac{1}{n_i} \quad \text{Var}[w_i] = \frac{1}{n_{i+1}}$$

This can only be satisfied if $n_i = n_{i+1}$, which is restrictive. We instead restrict

$$\frac{1}{\text{Var}[w]} = \frac{n_i + n_{i+1}}{2} \leftarrow \text{AVERAGE NUMBER OF UNITS.}$$

$$\therefore \text{Var}[w_i] = \frac{2}{n_i + n_{i+1}}$$

If weights are uniformly sampled from $[-a, a]$, the variance is given by

$$\begin{aligned} \text{Var}[w] &= E[w^2] = \int_{-a}^a w^2 \cdot \frac{1}{a - (-a)} dw \\ &= \int_{-a}^a \frac{w^2}{2a} dw \\ &= \left[\frac{w^3}{6a} \right]_{-a}^a = \frac{a^2}{6} + \frac{a^2}{6} \\ &= \frac{2a^2}{3}. \end{aligned}$$

Thus, passing in our bound

$$\frac{a^2}{3} = \frac{2}{n_i + n_{i+1}}$$

$$\Rightarrow a = \frac{\sqrt{6}}{\sqrt{n_i + n_{i+1}}}$$